
Future Improvement: Make E4 Evaluation Economically and Risk-Structurally Real

Problem summary

The current E4 evaluation pipeline is functionally complete but economically naive:

- Performance metrics (total_return ≈ 10%, Sharpe ≈ 345) are generated off an unrealistically smooth, low-volatility PnL path.
- Scenarios are generic ("All +10% / All −10%") and not tied to thesis drivers or disconfirmers.
- Risk metrics (VaR, max drawdown, max_weight_pct) are calculated, but they are not linked to the thesis narrative or factor structure.
- The LLM review sees the thesis text and the headline metrics but has no structured view of how drivers, disconfirmers, and exposures interact.

E5 and E6 deliberately **do not** solve these deficiencies – they add paper execution (E5) and UI/use-stage wiring (E6) on top of whatever E4 provides.

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1. Driver-linked scenario engine

Current behavior

- E4 emits a small set of generic scenarios:
  - "All −10%", "All +10%" with pnl_abs / pnl_pct.
- These scenarios don't correspond to any of the thesis' stated drivers (e.g. "inflation surprise up", "USD strength", "real yields rising/falling").

Why this is a problem

- The LLM correctly criticizes that it "can't see how drivers/disconfirmers quantitatively affect the portfolio".
- There's no way to test "what if the disconfirmer hits but not the primary driver" or to see asymmetric impacts.

What is not in E5/E6

- E5: introduces TradePlan and paper execution, but keeps sizing naive and doesn't define any driver-scenario machinery.
- E6: only surfaces whatever E4/E5 already compute; it doesn't add new scenario logic.

Future work

- Design a "driver-bound scenario engine" that:
    - Maps each driver/disconfirmer into one or more concrete scenario shocks on the underlying assets / factors.
    - Produces scenario P&L per thesis and at portfolio level.
    - Exposes a structured schema so both the UI and LLM review can talk in terms of "Driver D1 → Scenario S1 → ΔP&L".

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2. Realistic price paths and risk metrics

Current behavior

- The timeseries in the sample E4 output is an almost perfectly linear upward path (1.00 → ~1.10+) with tiny day-to-day changes.
- Volatility is ~1% annualized and Sharpe is >300, which is mathematically consistent with the path but economically absurd.
- VaR and max drawdown are computed off that synthetic path.

Why this is a problem

- Any downstream decision (E5 sizing, E6 UI, later risk engines) built on these numbers will be misleading – they reflect the toy path, not market reality.
- LLM critique cannot meaningfully comment on "risk vs reward" if the risk inputs are essentially fake.

What is not in E5/E6

- E5: assumes there is a PriceSource but does not constrain how realistic the underlying time series must be.

- E6: only displays the time series and risk metrics; it doesn't enforce realism or sanity checks.

Future work

- Plug in a concrete PriceSource (DB or TwelveData) for E4/E5 evaluation.
- Define minimum realism constraints:
  - Use actual historical returns for the thesis assets over the thesis horizon.
  - Enforce basic sanity checks (e.g. Sharpe, vol, and drawdowns within plausible ranges; flag or reject if not).
- Optionally add a "simulation mode" that still produces stochastic paths but with calibrated vol/correlation instead of a monotonic drift.

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3. Factor- and driver-aware risk diagnostics

Current behavior

- E4 emits a small risk_metrics blob:
  - max_weight_pct, VaR_95, and max_drawdown_pct.
- There's no breakdown by:
  - factor exposures (rates, FX, inflation, growth, etc.), or
  - drivers/disconfirmers as defined in the thesis.

Why this is a problem

- E4 can't answer "How much of this thesis is effectively a bet on real yields vs a USD bet vs pure gold beta?"
- LLM can only talk about concentration in asset space (e.g. "100% GLD"), not risk space.

What is not in E5/E6

- E5: talks about risk-aware sizing as a later engine, but doesn't specify factor-level diagnostics, and v1 sizing is explicitly naive.
- E6: shows whatever E4/E5 give it – no additional factor analysis.

Future work

- Introduce a factor model (even a simple 3–5 factor structure: real yields, DXY, equity beta, curve steepener, inflation).
- Compute and store:
    - factor betas at thesis level,
    - contribution to variance / VaR by factor,
    - concentration metrics in factor space.
- Feed this into:
    - E4 evaluation outputs,
    - E5 (later) risk-aware sizing engine,
    - E6 UI (e.g. "this thesis is 80% driven by real yields").

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4. Tighten LLM review ↔ quant coupling

Current behavior

- LLM review reads:
    - thesis text (drivers/disconfirmers + expression legs),
    - top-level metrics (total_return, Sharpe, VaR, scenarios).
- Critique, questions, and risk flags are high-level and mostly qualitative ("you're concentrated in GLD", "no scenario mapping").

Why this is a problem

- Review isn't truly "risk-aware" – it doesn't notice, for example:
    - contradictions between stated risk tolerance and realized vol/drawdown,
    - factor concentrations that conflict with the user's narrative,
    - scenario outcomes that invalidate the thesis.

What is not in E5/E6

- E5: defines how to turn thesis → TradePlan, then paper-execute; it doesn't prescribe a richer review protocol.
- E6: only displays the latest LLM review output.

Future work

- Define a richer, structured evaluation payload for the LLM that includes:
    - factor exposures,

    - driver-linked scenario results,
    - comparison of realized vs target risk (vol, drawdown, VaR).
- Update the E4 review prompt so the LLM must:
    - explicitly reconcile narrative vs metrics,
    - flag inconsistencies,
    - reference specific numbers/factors when raising risk flags.

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5. Bridge from naive sizing (E5 v1) to risk-aware sizing

Current behavior

- E4: only evaluates; no sizing.
- E5 v1: NaiveSizingEngine maps expression legs directly to TradeLeg size_pct (long-only, no vol/risk logic).

Why this is a problem

- Even once E5 is implemented, "approved" theses will produce TradePlans that ignore:
    - asset volatility,
    - correlation structure,
    - user- or portfolio-level risk limits.

What is not in E5/E6

- E5 explicitly states that risk-aware sizing is a later, pluggable engine, not part of v1.
- E6 just shows whatever E5 gives it.

Future work

- Design a RiskAwareSizingEngine that:
    - consumes E4 risk diagnostics (factor betas, VaR, driver scenarios),
    - respects per-thesis and portfolio-level risk constraints,
    - enforces concentration limits in both asset and factor space.
- Keep the TradePlan schema intact but replace the naive sizing logic with risk-aware sizing once the evaluation stack is mature.
